Tests of Significance for Large Samples

Statistics

Classification of Statistical Tests

- Sampling of Attributes
  - Test for proportion
  - Test for difference between proportions
- Sampling of Variables
  - Tests for mean, SD, correlation, etc.

Test for Proportion of Successes

- Used when data are classified into success and failure
- Sample proportion:

$$p = \frac{x}{n}, \quad q = 1 - p$$

- Population proportion estimated from sample

Standard Error of Proportion

$$S.E.(p) = \sqrt{\frac{pq}{n}}$$

- Measures reliability of sample proportion
- Smaller S.E. implies greater accuracy

Confidence Limits of Proportion

- 95% confidence limits:

$$p \pm 1.96 \times S.E.$$

- 99% confidence limits:

$$p \pm 2.58 \times S.E.$$

- Indicates probable range of population proportion

Illustration: Bad Apples Problem

- Sample size $n = 500$
- Bad apples $= 50$
- Sample proportion:

$$p = 0.1, \quad q = 0.9$$

- Standard Error:

$$S.E. = \sqrt{\frac{0.1 \times 0.9}{500}} = 0.013$$

Limits of Bad Apples Percentage

$$0.1 \pm 3(0.013)$$

- Lower limit $= 0.061$
- Upper limit $= 0.139$
- Percentage range:

$$6.1\% \text{ to } 13.9\%$$

Test for Difference Between Two Proportions

- Used to compare two population proportions
- Null Hypothesis:

$$H_0 : P_1 = P_2$$

- Difference assumed due to sampling fluctuations

Pooled Proportion

$$p = \frac{x_1 + x_2}{n_1 + n_2}$$

- Combines information from both samples
- Used in standard error calculation

Standard Error of Difference of Proportions

$$S.E.(p_1 - p_2) = \sqrt{pq\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$

- Basis of Z-test for proportions

Decision Rule

- $|Z| < 1.96$: Not significant (5%)
- $|Z| > 2.58$: Significant (1%)
- Determines acceptance or rejection of hypothesis

Illustration: Wheat Consumers

- Town A: $n_1 = 1000,\ p_1 = 0.4$
- Town B: $n_2 = 800,\ p_2 = 0.5$
- Pooled proportion:

$$p = \frac{400 + 400}{1800} = 0.444$$

- Z-value = 4.17

Conclusion of Illustration

- Calculated Z ¿ 2.58
- Difference is statistically significant
- Towns differ in wheat consumption

Tests of Significance for Large Samples

- Sample size greater than 30
- Sampling distribution approximately normal
- Sample values close to population values

Objectives of Large Sample Tests

- Compare observation with expectation
- Estimate population parameters
- Measure reliability of estimates

Standard Error of Mean

- Population SD known:

$$S.E.(\bar{x}) = \frac{\sigma}{\sqrt{n}}$$

- Population SD unknown:

$$S.E.(\bar{x}) = \frac{s}{\sqrt{n}}$$

Fiducial Limits of Mean

- 95% limits:

$$\bar{x} \pm 1.96 \frac{\sigma}{\sqrt{n}}$$

- 99% limits:

$$\bar{x} \pm 2.58 \frac{\sigma}{\sqrt{n}}$$

Illustration: Mean Height Test

- Sample mean $=$ 64 inches
- Population mean $=$ 67 inches
- $\sigma = 3, \ n = 100$
- S.E. $= 0.3$
- Difference $=$ 10 S.E.

Conclusion of Height Test

- Difference ¿ 1.96 S.E.
- Null hypothesis rejected
- Population mean is not 67 inches

Standard Errors of Other Statistics

- Median:

$$1.253\frac{\sigma}{\sqrt{n}}$$

- Quartiles:

$$1.36\frac{\sigma}{\sqrt{n}}$$

- Quartile Deviation:

$$0.787\frac{\sigma}{\sqrt{n}}$$

More Standard Errors

- Standard Deviation:
$$\frac{\sigma}{\sqrt{2n}}$$
- Variance:
$$\sigma^2\sqrt{\frac{2}{n}}$$
- Correlation coefficient:
$$\frac{1-r^2}{\sqrt{n}}$$
- Rank correlation:
$$\frac{1}{\sqrt{n-1}}$$

Conclusion

- Standard error is the foundation of inference
- Large sample tests rely on normal approximation
- Widely used in economics, commerce, and social sciences